

Virtual-to-Physical Surface Alignment and Refinement Techniques for Handwriting, Sketching, and Selection in XR

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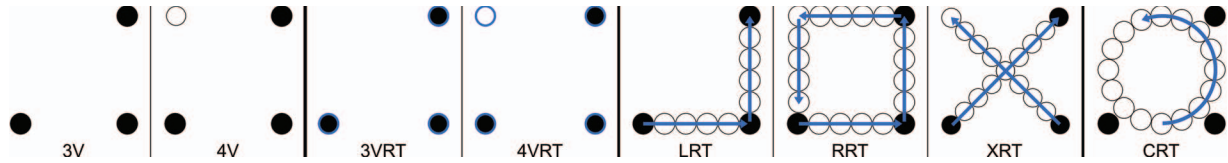


Figure 1: The figure shows the 3ViSuAI (3V) [14] alignment technique and our refinement techniques. 4ViSuAI (4V) extends 3V by adding a fourth reference point at the top left for a more uniform distribution of points over the surface. Since three single measurements can lead to inaccuracies, the 3V refinement technique (3VRT) and the 4V refinement technique (4VRT) record multiple measurements per reference point. Based on continuous motion, we consider the line refinement technique (LRT), the rectangular refinement technique (RRT), and the cross refinement technique (XRT). Finally, as a hybrid between discrete and continuous motion, we propose the circle refinement technique (CRT).

ABSTRACT

The alignment of virtual to physical surfaces is essential to improve symbolic input and selection in XR. Previous techniques optimized for efficiency can lead to inaccuracies. We investigate regression-based refinement techniques and introduce a surface accuracy evaluation. The results revealed that refinement techniques can highly improve surface accuracy and show that accuracy depends on the gesture shape and surface dimension. Our reference implementation and dataset are publicly available.

Keywords: Virtual Reality, Augmented Reality, Extended Reality, Plane, Surface, Calibration, Alignment, Passive Haptic Feedback.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms;

1 INTRODUCTION AND RELATED WORK

The alignment of virtual content to physical surfaces is essential to improve symbolic input and selection in XR. For example, sketching [3, 10, 12, 14], handwriting [9, 13, 14, 20], and virtual keyboard typing [1, 8] are important applications for future workspaces in virtual reality (VR) and augmented reality (AR), for short extended reality (XR). Physically aligned surfaces can also improve input accuracy and reduce user arms' fatigue [4, 14, 21].

Previous work has shown that a rectangular 2D surface can be aligned to a planar physical surface with only three points [14]. While this technique allows for fast execution, inaccuracies can arise due to the low number of physical reference points, which then may lower the reliable detection of interaction intent. Arora et al. [2] used a stroke-based approach to align the virtual surface. However, this approach only approximates the orientation and dimensions of the surface and is not determined exactly as in Kern et al. [14]. A faster but less flexible approach requires the placement of the VR controller on the physical table to determine the position of a virtual representative [16, 23, 25]. Even though the alignment with a single measure point can lead to reasonable results when high-precision

VR/AR controllers are used, it is also limited by their availability, and restricts flexibility in terms of surface dimension definition.

Consequently, there is a gap in research into surface alignment techniques that are both fast and lightweight, as well as highly accurate. In this work, we investigate different refinement techniques to extend the alignment technique 3ViSuAI [14] and introduce a continuous surface accuracy evaluation. For all refinement techniques, we use least squares regression analysis to compute the best-fit plane to a set of reference points. We follow the second research question of Kern et al. [14] regarding "How to align virtual 2D surfaces to arbitrary flat physical surfaces for handwriting and sketching input with XR styluses?" and ask the following research question: "How can the existing alignment technique 3ViSuAI be improved in its accuracy without significantly increasing its complexity?"

With our publicly available reference implementation and accuracy evaluation dataset, we encourage replication, reuse, and extension by researchers and developers, and provide a foundation for the development of novel alignment/refinement techniques and integration with other work.

2 CONCEPT

In the following, we present different refinement techniques and a continuous surface accuracy evaluation technique. The refinement techniques are designed to improve the accuracy of a physically aligned surface by recording numerous measure points. With these measures, a refinement plane is spanned using least squares regression analysis. Subsequently, the three reference points are refined by determining the closest points on the refinement plane, and the surface is aligned with 3V. Any physical point on an XR controller can be configured as the measurement reference. However, for precise measurements, we recommend using a dedicated or self-constructed controller stylus tip. We show the investigated alignment/refinement techniques in Fig. 1 and the most promising techniques in Fig. 3.

2.1 3ViSuAI (3V) and 4ViSuAI (4V)

Kern et al. [14] proposed 3ViSuAI (3V), a three point alignment technique to align a virtual rectangular 2D surface to a planar physical surface while directly determining its dimensions. This approach allows for fast execution on arbitrary planar surface orientations and was used for handwriting and sketching tasks. However, the authors also mentioned, that inaccuracies can arise due to the low number of three reference points, which then may lower the reliable detection of interaction intent. Therefore, we introduce 4ViSuAI (4V) to

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achieve a more uniform distribution of reference points by adding a fourth measurement at the top left of the surface. In 4V, the first three points are used to align the surface with 3V, and all four points are to refine the surface alignment.

2.2 3V and 4V Refinement Technique (3VRT, 4VRT)

While 4V can improve accuracy with a fourth reference point, surface alignment with only a single measurement per point can still result in a misaligned surface because of potential inaccuracies in the controller's position at the moment of acquisition. To address this issue, we propose the 3ViSuAI refinement technique (3VRT) and 4ViSuAI refinement technique (4VRT), which extend 3V and 4V with a continuous measurement per point instead of a single acquisition. For this, the user places the controller on the reference points and holds down a button to acquire multiple measurements. We then calculate the average position for each point, and use the first three points for 3V, and in 4V the fourth point for refinement.

2.3 Line Refinement Technique (LRT)

3VRT and 4VRT are promising approaches to reduce inaccuracies arising from controller jitter by averaging the positions of numerous measurements. However, these techniques use discrete recordings at predefined points rather than recording continuous motion as in handwriting and sketching. Our first continuous motion approach, the line refinement technique (LRT), combines 3V and the refinement process in a single gesture. With LRT, the user draws two connected perpendicular lines (reversed capital L) on the physical surface. The three 3V reference points are implicitly determined from the shape. We define the first and the last measured point of the gesture as the first and the third 3V reference point. The second point is defined by the largest distance from the first and third point. The measured points are used to span a refinement plane and determine the closest points of the 3V reference points to align the surface. LRT simplifies the process of surface spanning by a single gesture. However, recording points only along the bottom and right edges of the surface may result in lower accuracy in the upper left part.

2.4 Rectangle Refinement Technique (RRT)

For a more uniform distribution of reference points over the target surface, we introduce the rectangle refinement technique (RRT), an extension of LRT in which the user draws a rectangle with a continuous motion. RRT defines the first point for 3V by the first measured point of the rectangular gesture. The third point is the one that has the largest distance from the first point, and the second point is defined as the point between the first and third point with the largest distance from these points.

2.5 Cross Refinement Technique (XRT)

Instead of two perpendicular lines (LRT) or a closed rectangle (RRT), we also investigate the cross-refinement technique (XRT), in which the user draws two intersecting (crossing) lines across the surface. We take advantage of the continuous motion approach of LRT and RRT, and determine the 3V reference points implicitly from the gesture. While RRT uses the edges of the surface as the gesture shape, XRT focuses on a uniform distribution of the measure points across the target surface. The first and last measured point of the first line represents the first and the third point. In contrast to LRT and RRT, the second point is not determined by distance, but is the first point of the second line. Finally, we align the surface with 3V.

2.6 Circle Refinement Technique (CRT)

While LRT, RRT, and XRT use continuous motion to acquire reference points, this can also increase the difficulty of precisely matching predefined surface dimensions (e.g., DIN A1 or DIN A3).

Therefore, we propose the circle refinement technique (CRT), which adds a circular movement as a subsequent refinement procedure to 3V. This approach is inspired by the desk alignment technique presented by Meta in Horizon Workrooms, but in addition, uses the third measure point of 3V to determine the surface dimensions. CRT consists of two steps. At first, the user spans up the surface using 3V. Second, the user continuously records measure points in a fluent circular movement to define the refinement plane. CRT then determines the closest points on the refinement plane for the initial 3V points, and refines the surface.

2.7 Continuous Surface Accuracy Evaluation

Kern et al. [14] proposed a technique for evaluating surface accuracy using nine predefined discrete measurement positions. However, continuous motions, as in CRT, LRT, RRT, and XRT, more likely reflect handwriting and sketching on a surface than discrete points. Therefore, we propose a continuous motion accuracy evaluation in which a rectangle is traced on the spanned surface to record measure points. The techniques are shown in Fig. 2.

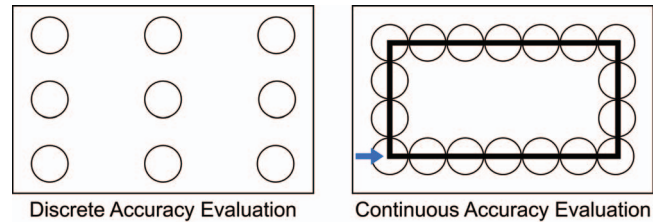


Figure 2: The figure shows the discrete accuracy evaluation proposed by Kern et al. [14] and our continuous surface accuracy evaluation approach. Instead of measuring at nine predefined positions, we propose to trace a rectangle that is visualized on the spanned surface.

2.8 System Control

A controller button or pressure sensitive tip (Logitech VR Ink or Meta Quest Touch Pro Controller) can be used to control the recording sequence. When using buttons for system control, previous work suggest placing the input control on the opposite controller to avoid inaccuracies due to unplanned movements caused by button presses [14, 19, 24] (so-called Heisenberg Effect [7]). In this work, we used the Meta Quest Pro Touch controller's stylus tip as the reference point and used the left controller buttons for input control.

3 REFERENCE IMPLEMENTATION

We used Unity 2020.3 LTS and the Oculus Integration SDK¹ with support for the Meta Quest Pro, the Meta Quest 2, and the Oculus Rift S. Other devices can be integrated with Unity's XR Plugin Framework². Detailed information can be found in the publicly available reference implementation. We created a new Unity project and copied the 3ViSuAI alignment technique from the official Off-The-Shelf-Stylus repository (see [14] for the link). We then added our implementations of the refinement techniques. We developed discrete and continuous recording features to capture measure points. To determine the refinement plane and evaluate the continuous surface accuracy, we used the least squares fitting implementation of MathUtilities³ and Unity's Plane.ClosestPointOnPlane⁴ feature. The visual representation of our reference implementation is shown in Fig. 3. Kern et al. [14] used a self-constructed stylus tip attached to the bottom of the controller and based their stylus calibration

¹<https://developer.oculus.com/unity/>

²<https://docs.unity3d.com/Manual/XRPluginArchitecture.html>

³<https://github.com/zalo/MathUtilities>

⁴<https://docs.unity3d.com/ScriptReference/Plane>

technique on an open-source solution (CalibrateTooltip⁵ by Anthony Steed). Other authors presented XR stylus tips using 3D printing (e.g., [6, 11, 15, 19]). Since the Meta Quest Pro Touch Controller includes a stylus tip by default, a self-constructed or 3D-printed stylus tip was not required for this work. However, a stylus tip is not mandatory, but it may increase the accuracy of the recorded points.

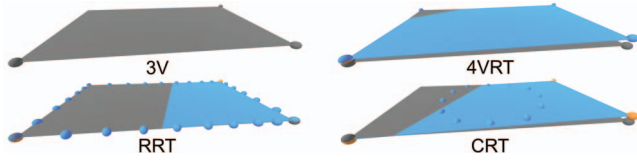


Figure 3: The figure shows 3V [14] and three selected refinement techniques (3VRT, RRT, and CRT). Gray spheres represent the three points for alignment with 3ViSuAI, blue spheres an arbitrary number of measured points for refinement, and orange spheres the realigned 3V reference points. Gray areas show the surfaces aligned with 3V, and the blue areas the surfaces refined.

4 TECHNICAL EVALUATION

We conducted an initial evaluation with two XR experts to get an idea of the usability and accuracy of 3ViSuAI and the improvement of our refinement techniques. The results are shown in Table 1. We used the Meta Quest Pro with Meta Quest Link, and a laptop with Intel Core i7-10875H, 16 GB DDR4, and NVIDIA GeForce RTX 2070 SUPER to avoid possible performance issues. A quick performance test showed that the number of recorded points and visualizations should have no effect on performance. The latter can be removed in future applications. Since surface dimensions may influence the accuracy, we examined two different sizes (DIN A3 297 x 420 mm and DIN A1 594 x 841 mm). The experts used 4VRT, LRT, RRT, XRT, and CRT five times for both surface dimensions and performed the continuous surface accuracy evaluation. We recorded 100 measurements per reference point for 4VRT, and one measurement every 5 mm for LRT, RRT, XRT, and CRT. The results for 3V, 4V, and 3VRT were generated from the 4VRT recordings for better comparability. The data was stored separately in CSV files for statistical analyses. Our dataset is publicly available for reuse, replication, and development of new techniques.

5 RESULTS

We calculated the average accuracy in millimeters for each technique and surface dimension (see Fig. 4 and Table 1). Further, we defined four areas equally distributed over the surface and calculated the average local accuracies (see Fig. 5). Our initial technical evaluation with two experts shows that enhancing 3ViSuAI with refinement techniques can greatly improve surface accuracy. In general, CRT and RRT achieved best results for both DIN A3 and DIN A1. Compared to 3V, the accuracy improvements with CRT and RRT of 59% (DIN A3) and for CRT 35% (DIN A1) are particularly noteworthy. Our results also show that a fourth reference point in the upper left corner can considerably improve the original 3V alignment. In contrast, there was only a small difference between single measurements (3V, 4V) and multiple measurements (3VRT, 4VRT) per reference point. The fine-grained analysis (see Fig. 5) confirms that each refinement technique has different effects on accuracy in individual areas of the spanned surface. While the uniform distribution of reference points in RRT, XRT and CRT is also reflected by similarly colored areas, the results for 3V, 3VRT and LRT show higher inaccuracy in areas without refinement points. Moreover, our results indicate that the larger the surface dimensions, the more noticeable the accuracy discrepancy.

⁵<https://github.com/anthonystead/CalibrateTooltip>

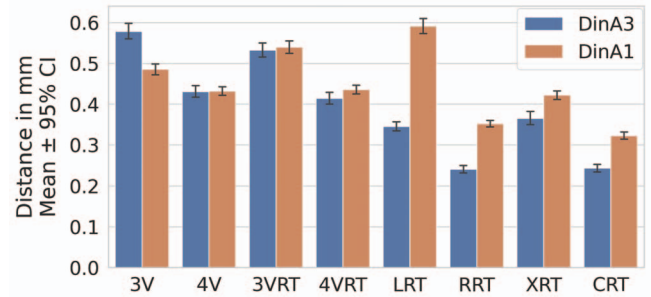


Figure 4: The chart shows the results of the continuous accuracy evaluation in millimeters (mm), averaged over all measure points.

6 EXAMPLE USE CASE: THE FUTURE XR WORKSPACE

In future XR workspaces, physically aligned virtual surfaces will play an essential role for productive and collaborative work. Our techniques enable the alignment/refinement of 2D surfaces to flat physical surfaces in arbitrary dimensions and orientations. Virtual surfaces are the basis for several use cases that highlight the importance and advantages of our techniques. For example, virtual surfaces can be used to align a regular web browser to a physical desk for handwriting and sketching using productivity tools such as Miro, Notion, and Microsoft Office 365. For meetings, collaborative or creative work, virtual surfaces can be aligned with vertical physical surfaces for whiteboard or wall interaction. Our techniques can also be used to align pure virtual objects (e.g., a virtual keyboard) or virtual replicas/digital twins of real objects, such as physical keyboards and laptops. For this, three/four points on the physical keyboard are defined and recorded by the user (e.g., the points between the adjacent keys SDX, JKM, 89I and 34E). Likewise, the alignment of physical room replicas is possible. Once a second user records the same physical reference points, we are able to align the different coordinate systems with sub-millimeter accuracy, enabling collaborative work in co-located environments. Finally, the broad compatibility of our techniques with arbitrary XR devices with controllers or hand tracking frees users from platform-specific implementations such as scene understanding or spatial anchoring.

7 DISCUSSION

The results of our technical evaluation show that our refinement techniques can further improve the accuracy of physically aligned virtual surfaces with a minimum of effort. We show that accuracy depends on the gesture shape and on the surface dimensions, potentially requiring the support of multiple techniques. In this work, CRT and RRT achieved best results for both DIN A3 and DIN A1. Compared to 3V and 3VRT, adding a fourth reference point in 4V and 4VRT showed higher accuracy, but still lower accuracy than other techniques (RRT, XRT, and CRT). We also observed that the Meta Quest Pro controllers increasingly jittered when placed in a fixed position on the table, which could explain similar results for 3V/3VRT and 4V/4VRT, instead of an expected accuracy improvement.

For the Oculus Rift S and DIN A4 surfaces (297 x 210 mm), Kern et al. [14] reported an average accuracy of 0.60 mm (SD = 0.32 mm). Compared to the Meta Quest Pro with Touch Pro controllers, our results indicate that surface accuracy may also be highly dependent on the overall accuracy of the XR device and controllers. We may also expect lower accuracies with other XR controllers/trackers like the HTC Vive Pro Controller [5, 22] and HTC Vive Tracker [18, 22], or hand tracking based solutions [8, 13, 17]. Nevertheless, the technical evaluation revealed first insights into the sub-millimeter tracking accuracy of the Meta Quest Pro Touch controllers with novel inside-out self-tracking.

Table 1: The table shows the results of the continuous accuracy evaluation in millimeters (mm), averaged over all measure points.

Dimensions	DIN A3 (297 x 420mm)								DIN A1 (594 x 841mm)							
	3V	4V	3VRT	4VRT	LRT	RRT	XRT	CRT	3V	4V	3VRT	4VRT	LRT	RRT	XRT	CRT
Mean	0.58	0.43	0.53	0.41	0.35	0.24	0.37	0.24	0.49	0.43	0.54	0.44	0.59	0.35	0.42	0.32
SD	0.41	0.32	0.37	0.31	0.23	0.20	0.34	0.20	0.40	0.34	0.43	0.35	0.40	0.24	0.31	0.26

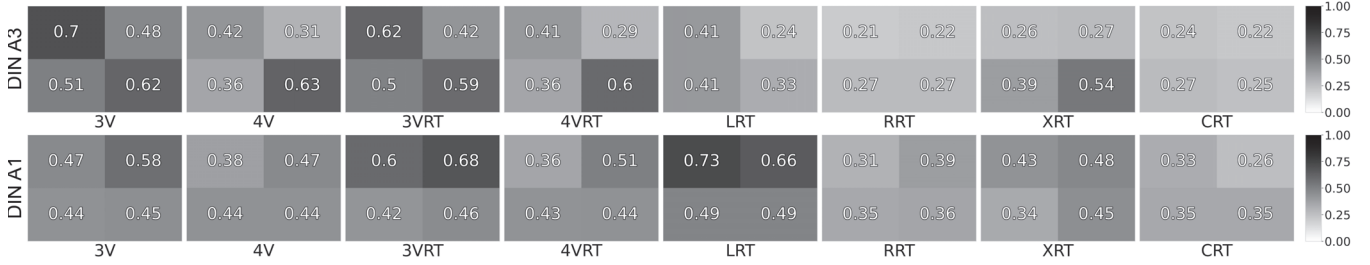


Figure 5: The figure shows a detailed insight into the results of our continuous surface accuracy evaluation for alignment and refinement techniques. We have divided each surface into four areas and report the averaged absolute distances in millimeters (mm).

8 CONCLUSION

The alignment of virtual to physical surfaces is essential to improve symbolic input and selection in XR, e.g., for handwriting and sketching or selecting keys on a virtual keyboard. In addition, physically aligned surfaces can improve input accuracy and reduce user arms' fatigue. In line with previous work, our results emphasize the importance of error-tolerant surface alignment and refinement techniques, as well as a possible impact of XR device tracking quality on surface accuracy. Considering possible inaccuracies, approaches to use arbitrary XR controllers as XR styluses are promising and enable the development of reusable software for handwriting, sketching, and selection in future XR workspaces. Besides the surface accuracy, execution speed is also an important criterion for usability, user experience and task load. While CRT and RRT have the best accuracy, they are also more complex and require the longest execution time.

The question arises which technique should be used? It depends. Our results suggest that continuous motion approaches such as CRT and RRT can achieve higher accuracy for large surface dimensions on flat physical surfaces (e.g. DIN A1), compared to discrete techniques (3VRT, 4VRT). However, for aligning the virtual replica of a physical keyboard using defined positions between keys, 4VRT may be more suitable. Therefore, the application scenario and applicability should be considered when selecting a technique.

We would like to encourage developers and researchers to integrate surface alignment/refinement techniques into their projects and to perform user evaluations. Therefore, we have made our reference implementation and accuracy evaluation dataset publicly available for reuse, replication, and extension (<https://go.uni-wue.de/alignment-and-refinement-techniques>).

9 FUTURE WORK AND LIMITATIONS

Based on our initial technical evaluation, we plan to conduct a comprehensive user evaluation following the evaluation suite of Kern et al. [14] including accuracy measures, handwriting, and sketching. In addition to subjective questionnaires, objective measures should also evaluate accuracy, execution time, and overall system performance. Besides structured approaches with predefined gestures for the refinement technique, we would also like to investigate unstructured movements such as hatching. Continuous refinement while writing and sketching may also be promising. Other XR devices, like the HTC Vive Focus 3 and Pico 4, will also be supported in our reference implementation. With our public dataset, we will develop and evaluate new alignment and refinement techniques.

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